Locking Away Prosperity? Evaluating the Economic Impact of the Key to NYC Program

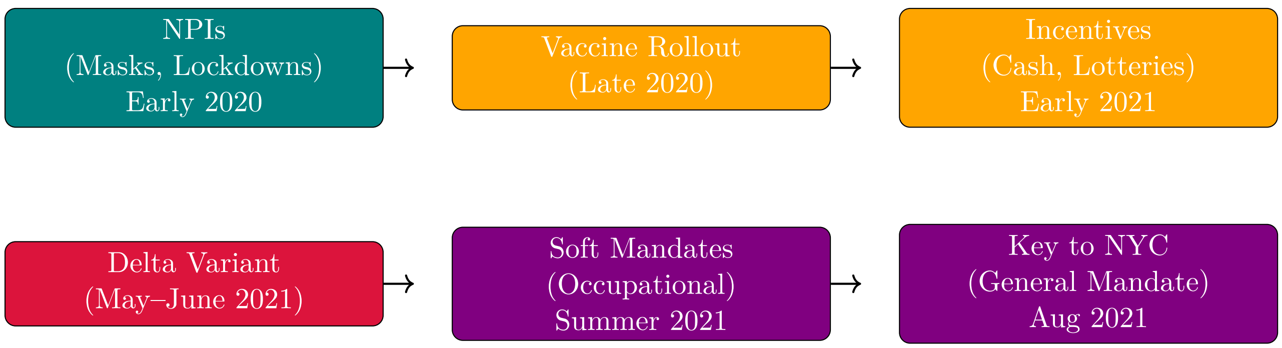
1 Introduction

The COVID-19 pandemic triggered a global wave of non-pharmaceutical interventions (NPIs) as governments acted swiftly to limit viral transmission. Much early research focused on the public health rationale for these policies—especially the importance of early action to contain exponential spread. For instance, Friedson et al. [2021] and Yang [2021] use synthetic control methods to evaluate lockdowns in California and Wuhan, emphasizing the critical role of timing. Similarly, Bayat et al. [2020] estimate that New York City could have reduced COVID-19 deaths by over 80% had its first lockdown occurred just ten days earlier. Subsequent work examined the economic consequences of such interventions. While NPIs helped flatten the curve, they also raised concerns about business closures, layoffs, and lasting economic scarring. Studies like Gibson and Sun [2023] and Lee and Yang [2022] document labor market costs of lockdowns, while Ke and Hsiao [2022] show that although strict NPIs incur short-term economic losses, those effects may be temporary. Vaccine mandates emerged as a less restrictive (but still controversial) policy tool during later phases of the pandemic. Unlike lockdowns, mandates aimed to curb transmission without shutting down businesses. Nevertheless, critics argued that mandates could hamper fragile economic recoveries by alienating customers or exacerbating staffing shortages in consumer-facing sectors like restaurants. Supporters countered that mandates would improve public safety and restore consumer confidence. These competing narratives came to a head with New York City's *Key to NYC* program—implemented in August 2021—the first large-scale vaccine mandate for indoor commercial activity in the U.S. [Towey, 2021, Rogers, 2021, Bardosh et al., 2022].

This paper evaluates the causal effect of the Key to NYC mandate on employment in New York City's full-service restaurant sector—the segment most directly affected by the policy. While prior research has analyzed mandates' effects on vaccination rates and health outcomes [Cohn et al., 2022, Acharya and Dhakal, 2021, Lang et al., 2023], relatively little is known about their labor market consequences, particularly in industries that rely heavily on face-to-face interaction. This study helps fill that gap. Because randomized experiments are infeasible, this paper adopts a synthetic control approach to estimate the counterfactual trajectory of employment absent the mandate. Using MSA-level monthly data from the Federal Reserve Bank of St. Louis, I construct a synthetic control for NYC's year-over-year growth in full-service restaurant employment. The donor pool consists of 29 MSAs that did not implement similar mandates during the same period.[1]

The rest of the paper is organized as follows. Section 2 provides background on the evolution of pandemic interventions and the Key to NYC program. Section 3 describes the employment data and outcome construction. Section 4 outlines the empirical strategy, including the synthetic control methodology. Section 5 presents the main findings and robustness checks. Section 6 interprets the results and discusses limitations. Section 7 concludes.

2 Policy Background



Timeline of major pandemic policy phases

2.1 NPIs Before Vaccines

NPIs had aimed to reduce transmission by limiting contact or promoting respiratory barriers. The frst and most famous of these interventions was the 2020 lockdwn in Wuhan, China, where local officials went as far as to forcibly seal people in their homes to reduce the possibility to spread the virus; similar actions were taken in Europe and India. Even in the United States, states that could effectively seal their borders, such as Hawaii, did so, with Hawaii mandating two-week quarantines to visitors, which effectively amounted to a tourism ban, Mask mandates were widely adopted as welldue to their low cost and direct mechanism of action [Hansen and Mano, 2023, Chernozhukov et al., 2021, Mitze et al., 2020]. While not as severe as lockdowns, the idea of mask mandates was to allow for some freedom of movement while minimizing the chance to spread the virus further. While both sets of policies were effective, they provoked political and economic backlash due to their disruptive nature.

2.2 The First Vaccine Policies

The arrival of vaccines in late 2020 ushered in a new phase of pandemic governance, giving jurisdictions a wider array of policy options to employ. Now, instead of lockdowns, governments finally had pharmaceutical interventions in their toolkit. For the first few months, it worked: when vaccines were released, initial uptake was high, particularly among older

[1] For example, 'SMU11000007072251101SA' (Washington, DC) or 'SMU08197407072251101SA' (Denver MSA).

adults and (often mandated for) healthcare workers. However, as eligibility expanded in early 2021, demand slowed amid misinformation, mistrust, and logistical barriers. In response, policymakers deployed incentive-based strategies to encourage vaccination [Barber and West, 2022, Robertson et al., 2021]. These included direct cash payments, vaccine lotteries (e.g., Ohio's "Vax-a-Million" from May to June [Fuller et al., 2022]), and smaller symbolic give-aways. One paper dubbed this the era of "donuts, drugs, booze, and guns," reflecting the extraordinary lengths to which governments went to boost uptake through rewards rather than requirements [Tinari and Riva, 2021].

Despite their creativity, most incentives had modest or short-lived effects [Saban et al., 2021, Walkey et al., 2021, Khazanov et al., 2023]. Many were poorly targeted, administratively complex, or insufficient to overcome deeper resistance. The emergence of the highly transmissible Delta variant in mid-2021 raised the stakes. Initially identified in India, Delta reached the United States by May and accounted for the majority of new cases by June. Unlike in 2020, officials now had vaccines at scale—but not enough vaccinated people. Given a new more contagious strain, soft mandates emerged, targeting workers in high-risk occupations such as healthcare and education. These policies were not general mandates but did make vaccination a condition of continued employment [American Medical Association et al., 2021]. Over the summer, local governments and organizations expanded this approach [American Medical Association et al., 2021, Al Jazeera, 2021]. By August 2021, New York City introduced the Key to NYC program—one of the first large-scale vaccine mandates in the U.S. applied to the general public. Unlike prior efforts that relied on encouragement, Key to NYC made access to indoor dining, fitness, and entertainment conditional on vaccination status. It represented a new policy mechanism: exclusion rather than incentive.

2.3 The First Vaccine Mandates

Per the above, vaccine mandates could influence employment in full-service restaurants through both supply- and demand-side channels, with theoretically ambiguous net effects. On the supply side, some workers may exit the labor force or shift sectors if unwilling to comply, especially in lower-wage service jobs where vaccine hesitancy may be higher. However, others may comply with the mandate precisely *because* they wish to keep their jobs, choosing continued employment over personal objection. That is, the employment effect may hinge less on individual preferences in isolation and more on the strength of the economic incentives at stake. Employers, meanwhile, may face frictional costs from enforcing mandates or accommodating staffing turnover. On the demand side, mandates may increase consumer confidence by signaling a safer environment, thereby encouraging more patrons to return to indoor dining. For a sector like full-service restaurants, where both interpersonal contact and discretionary spending are central, mandates could plausibly stimulate demand even as they introduce short-run supply constraints. Whether employment rises or falls in net depends on which margin dominates. In short, this analysis seeks to quantify the net effect of the mandate on restaurant employment, capturing the balance between these potentially opposing forces.

3 Dataset

We use monthly MSA-level employment series from the Federal Reserve Bank of St. Louis (FRED) and the Bureau of Labor Statistics, spanning January 1991 through August 2022, yielding $T = 380$ monthly observations per unit. Our analysis focuses on three related outcome sets: (1) full-service restaurant employment for 30 MSAs including New York City as the treated unit with 29 donor MSAs; (2) broader leisure industry employment at the MSA level for approximately 255 MSAs from 1991 to December 2022. For the broader leisure employment analysis, New York City (FRED code SMU36935617072251101SA) is the treated unit, with the remaining MSAs as the donor pool. The broader leisure industry MSA analysis extends this framework to approximately 255 MSAs observed through December 2022 (FRED code SMU36000007072200001SA), providing a larger donor pool and outcome set.

Let e_{jt} denote employment in unit j at month t. For both the full-service restaurant analysis and broader leisure market analysis, the outcome is cobstructed as the year-over-year growth rate in percentage points,

$$y_{jt} = 100 \cdot \frac{e_{jt} - e_{j,t-12}}{e_{j,t-12}}, \qquad j = 1, \dots, N, \ t = 13, \dots, T.$$

of the smoothed and seasonally adjusted values. Outcomes are collected into vectors $\mathbf{y}_j = (y_{j1}, \dots, y_{jT})^\top$, and the donor pool is defined as $\mathcal{N}_0 = \mathcal{N} \setminus \{1\}$ with $N_0 = 29$. The *Key to NYC* policy, implemented in August 2021, designates July 2021 as the last pre-treatment month T_0, with the treatment indicator defined by

$$d_{jt} = \mathbf{1}\{j = 1 \wedge t > T_0\}.$$

Observed outcomes satisfy

$$y_{jt} = (1 - d_{jt}) y_{jt}(0) + d_{jt} y_{jt}(1),$$

with causal effects $\tau_{jt} = y_{jt}(1) - y_{jt}(0)$ and post-treatment average treatment effect on the treated (ATT)

$$\text{ATT} = \frac{1}{|\mathcal{T}_2|} \sum_{t \in \mathcal{T}_2} \tau_{1t},$$

where $\mathcal{T}_2 = \{t > T_0\}$.

%## CSA analysis

%To capture broader regional economic dynamics, we aggregate MSA-level leisure industry data into 154 CSAs covering the same period. Let $\mathcal{I}$ denote the set of all MSAs and $\mathcal{C}$ the set of all CSAs. Each CSA $c \in \mathcal{C}$ is defined as the union of its constituent MSAs, i.e.,

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%where $\mathcal{I}_c \subseteq \mathcal{I}$ is the subset of MSAs nested within CSA c. Equivalently, the sets $\{\mathcal{I}_c : c \in \mathcal{C}\}$ form a partition of $\mathcal{I}$ such that %

%We apply a Törnqvist index to compute CSA-level year-over-year growth rates that account for changing relative employment shares of constituent MSAs. For CSA c at month t, the growth rate is

%

%where weights $w_{i,t}$ are the average of MSA i's employment share in CSA c at months t and $t-12$:

%

%This weighted log-growth aggregation mitigates distortions from simple averaging and reflects the evolving composition of CSAs over time.

4 Econometric Methodology

To estimate the counterfactual employment trajectory for NYC's full-service restaurant sector and the broader employment trends in the leisure labor market, I employ a synthetic control approach that hybridizes Forward Selection SCM (FSCM) and the Augmented SCM (ASCM) estimators. The classical Synthetic Control Method (SCM) solves

$$\mathbf{w}^{\text{SCM}} = \underset{\mathbf{w} \in \mathcal{W}_{\text{conv}}}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \mathbf{w} \right\|_2^2,$$

where

$$\mathcal{W}_{\text{conv}} = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{N_0} \;\middle|\; \mathbf{1}^\top \mathbf{w} = 1 \right\}$$

is the probability simplex. This finds the weights $\mathbf{w}^{\text{SCM}}$ that best reconstruct the treated unit's pre-treatment outcomes as a convex combination of donor outcomes.

4.1 Forward Selection SCM

FSCM constructs the synthetic control iteratively by adding donor units one at a time. Starting with an empty selected donor set $\mathcal{S} = \emptyset$, at each iteration the algorithm evaluates each candidate donor $j \in \mathcal{N}_0 \setminus \mathcal{S}$ by forming an expanded set $\mathcal{S}' = \mathcal{S} \cup \{j\}$. For each $\mathcal{S}'$, it solves the restricted SCM problem over the donor submatrix $\mathbf{Y}_0^{\mathcal{S}'}$, consisting of columns corresponding to units in $\mathcal{S}'$:

$$\mathbf{w}^*_{\mathcal{S}'} = \underset{\mathbf{w} \in \mathcal{W}_{\text{conv}}(\mathcal{S}')}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1, \mathcal{S}'} \mathbf{w} \right\|_2^2,$$

where

$$\mathcal{W}_{\text{conv}}(\mathcal{S}') = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{|\mathcal{S}'|} \mid \mathbf{1}^\top \mathbf{w} = 1 \right\}.$$

The donor selected at iteration k is

$$j^* = \underset{j \in \mathcal{N}_0 \setminus \mathcal{S}}{\operatorname{argmin}} \min_{\mathbf{w} \in \mathcal{W}_{\text{conv}}(\mathcal{S} \cup \{j\})} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1, \mathcal{S} \cup \{j\}} \mathbf{w} \right\|_2^2,$$

and $\mathcal{S}$ is updated as

$$\mathcal{S} \leftarrow \mathcal{S} \cup \{j^*\}.$$

This forward selection continues until a stopping criterion is met, such as reaching a maximum number of donors, a minimum improvement threshold, or an information criterion like modified BIC.

4.2 The Augmented Synthetic Control Estimator

Building on the baseline SCM, ASCM relaxes the convex weight restriction by allowing affine weights and adds a regularization penalty that shrinks the solution towards a reference weight vector $\mathbf{w}_0$, typically the SCM or FSCM solution. The augmented objective is

$$\mathbf{w}^{\text{aug}} = \underset{\mathbf{w} \in \mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \left\| \mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1} \mathbf{w} \right\|_2^2 + \lambda \left\| \mathbf{w} - \mathbf{w}_0 \right\|_2^2,$$

where $\lambda \geq 0$ controls the penalty strength and

$$\mathcal{W}_{\text{aff}} = \left\{ \mathbf{w} \in \mathbb{R}^{N_0} \mid \mathbf{1}^\top \mathbf{w} = 1 \right\}$$

is the affine subspace of weights summing to one, but not restricted to be non-negative. The intuition is that when $\mathbf{w}_0$ (e.g., FSCM weights) provides a good fit, λ will be large to maintain interpretability; when fit needs improvement, λ can be small to allow extrapolation beyond the convex hull.

4.3 Forward Augmented Synthetic Control

FASC integrates the interpretability of FSCM with the flexibility of ASCM by first performing forward selection to obtain $\mathbf{w}^{\text{FSCM}}$, then solving

$$\mathbf{w}^{\text{FASC}} = \underset{\mathbf{w}\in\mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \left\|\mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1}\mathbf{w}\right\|_2^2 + \lambda \left\|\mathbf{w} - \mathbf{w}^{\text{FSCM}}\right\|_2^2,$$

where $\lambda \geq 0$ balances fidelity to the sparse forward-selected weights and pre-treatment fit.

Large λ yields a solution close to $\mathbf{w}^{\text{FSCM}}$, preserving sparsity and interpretability, whereas small λ prioritizes fit, allowing positive or negative weights outside the original selected donors.

4.3.1 Cross-Validation for λ

The hyperparameter λ is chosen by time-split cross-validation over the pre-treatment period. The training set (first half of pre-treatment) is used to estimate

$$\mathbf{w}^{\text{FASC}}(\lambda) = \underset{\mathbf{w}\in\mathcal{W}_{\text{aff}}}{\operatorname{argmin}} \left\|\mathbf{y}_{\text{train}} - \mathbf{Y}_{0,\text{train}}\mathbf{w}\right\|_2^2 + \lambda \left\|\mathbf{w} - \mathbf{w}^{\text{FSCM}}\right\|_2^2,$$

while the validation set (second half of pre-treatment) evaluates out-of-sample error via

$$\text{CV-RMSE}(\lambda) = \left\|\mathbf{y}_{\text{val}} - \mathbf{Y}_{0,\text{val}}\mathbf{w}^{\text{FASC}}(\lambda)\right\|_2.$$

The λ minimizing CV-RMSE balances interpretability and fit. Bayesian optimization over $\log_{10}(\lambda) \in [-2, 3]$ with Gaussian Process surrogates and Expected Improvement acquisition efficiently explores this space. Since $\mathbf{w}^{\text{FSCM}}$ is feasible in the FASC optimization, the in-sample Mean Squared Error (MSE) satisfies

$$\text{MSE}^{\text{FASC}} = \frac{1}{T_0}\left\|\mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1}\mathbf{w}^{\text{FASC}}\right\|_2^2 \leq \frac{1}{T_0}\left\|\mathbf{y}_1^{\mathcal{T}_1} - \mathbf{Y}_0^{\mathcal{T}_1}\mathbf{w}^{\text{FSCM}}\right\|_2^2 = \text{MSE}^{\text{FSCM}},$$

ensuring FASC's fit is never worse and typically better than FSCM.

5 Results

6 Discussion

7 Conclusion

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